

Notes: Geng, Liu, and Pan (WP, 2025)
The Government-Led Credit Cycle in China's LGFV Bonds and the Real Effects

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1 Big picture

- **Question.** How did China’s sequence of policies toward local government hidden debt change the pricing of *local government financing vehicle bonds (LGFV)*,¹ and what were the downstream real effects on local commercial banks?
- **Why LGFV bonds are the right asset.** LGFV bonds are unofficial off-budget obligations tied closely to local governments. They are effectively the “junior” claims of local governments, so they should be especially sensitive to any change in perceived central-government support.
- **Main empirical fact.** The paper identifies a **government-led credit cycle** in LGFV bonds. The spread gap between risky and healthy cities goes through three divergence waves and then a sharp convergence. The timing lines up with four major policy moments: Directive No.43 in 2014, the New Asset-Management Regulation in 2018, stricter anti-bailout credit regulation beginning in late 2020, and the 2023 Debt Resolution package.
- **Core mechanism.** The credit cycle is not mainly about raw regional fundamentals getting dramatically worse in some places and better in others. The paper argues that the main driver is changing **beliefs about central-government bailout probability**. When investors think Beijing is less likely to rescue risky local debts, spreads in risky cities become much more sensitive to fundamentals. When they think bailout odds rise again, those spreads compress.
- **Quantitative decomposition.** From January 2017 to the divergence peak in December 2022, the increase in the risky-minus-healthy spread gap is decomposed into about 28% fundamentals, about 7% liquidity, and about 65% bailout repricing.
- **Implied bailout path.** One calibration in the paper says that if the bailout probability is 90% in January 2017, then it falls to 71% by December 2019, to 36% by December 2022, and then recovers to 86% by July 2025 after Debt Resolution.
- **Main spillover result.** Debt Resolution narrows LGFV spreads in high-risk places and this relief spills over into lower subordinated-bond spreads for local banks, especially because LGFVs are major borrowers from those banks.
- **Main real-effects result.** The same policy that improves bank solvency also hurts local banks in risky cities by lowering profitability, shrinking their market share, and raising their probability of consolidation. Those banks respond by relying less on traditional lending and more on financial investments.
- **Big takeaway.** Government debt policy can generate a credit cycle even without a classic private-sector boom-bust. In this setting, the cycle is driven by state policy and shifting beliefs about which liabilities the central government will ultimately stand behind.

2 Data and measurement (key objects)

¹In China, a LGFV is an investment company formed by local governments to raise funds, usually by borrowing money to finance real estate development and other local infrastructure projects. (Wikipedia)

2.1 Policy timeline and how the sample is organized

The paper studies a long policy sequence rather than a single event. The sample is divided into five phases using four policy breakpoints:

- pre-Directive No.43, before October 2014;
- post-Directive No.43, October 2014 to March 2018;
- post-New Regulation, April 2018 to October 2020;
- post-Strict Regulation, November 2020 to June 2023;
- post-Debt Resolution, from July 2023 onward.

Interpretation. The paper's object of interest is the *evolution* of credit pricing across these policy regimes. The five-phase split is the basic time structure behind the whole note.

2.2 LGFV bond sample

The main data come from Wind and cover traded, fixed-rate LGFV bonds from January 2011 to July 2025. The paper keeps the three major public LGFV bond types:

- medium-term notes in the interbank market,
- exchange-traded corporate bonds,
- enterprise bonds that can trade in either market.

It drops private placements and bonds with maturity below one year, and it focuses on traded bonds so that market pricing is directly observed.

The paper also keeps some issuers that recently asked to be removed from Wind's LGFV list after claiming they had become ordinary enterprises. The authors treat those transitions cautiously and continue to view them as LGFVs.

Interpretation. The paper is not using accounting classifications mechanically. It is trying to measure the market for obligations that investors still think are tied to local governments.

2.3 How credit spreads are measured

For LGFV bonds, the main dependent variable is **credit spread**, defined as yield-to-maturity minus the Chinese Development Bank (CDB) yield of the same maturity, where the CDB curve is obtained by linear interpolation.

The paper uses the last transaction price of each month to compute the monthly yield.

Interpretation. Using the CDB curve gives a quasi-risk-free benchmark inside China's own bond market. So the spread is the market price of everything beyond that benchmark: default risk, liquidity, and any implicit-government-support premium.

2.4 How risky versus healthy cities are defined

Each month, the paper sorts cities into terciles using the city-level issue-size-weighted average spread of 1-to-5-year LGFV bonds over the previous 12 months. Cities in the top tercile are called **risky**; cities in the bottom tercile are the healthy comparison group.

The continuous version of this idea is **CityRiskyMeasure**, the pre-policy city-level average spread used in the bank regressions around Debt Resolution.

Interpretation. The paper is not saying a city is intrinsically risky forever. It uses lagged market prices to classify how exposed a city looks at each point in time.

2.5 Bond-level controls and liquidity measures

The LGFV-bond regressions control for standard bond characteristics:

- rating,
- maturity,
- bond age,
- issue size,
- exchange-trading indicator,
- callable and puttable features.

The paper also builds two price-based liquidity measures:

- the **Amihud ratio**, which captures price impact per unit of trading amount;
- the **TwoDayHighLow** measure of Abdi and Ranaldo, which infers effective bid-ask spreads when quote data are largely missing.

Interpretation. These variables matter because a wider spread on an LGFV bond can reflect not only default or bailout risk, but also higher trading frictions.

2.6 LGFV and regional fundamentals

At the issuer level, the paper tracks firm size, profitability, and leverage. Over time, LGFVs in the sample become less profitable and more levered.

At the provincial level, the paper uses four fundamental proxies:

- debt-to-GDP ratio,
- log GDP,
- fiscal self-sufficiency ratio (budget revenue divided by expenditure),
- average municipal-bond spread over the prior 12 months.

These variables are lagged to avoid forward-looking contamination.

Interpretation. The paper wants a broad set of fundamentals that can proxy both repayment capacity and expected loss in no-bailout states. The municipal spread variable is especially useful because it is itself a market-based summary of local fiscal stress.

2.7 Municipal bonds and local bank bonds

To study spillovers, the paper also collects:

- provincial municipal bonds, starting when independently issued municipal bonds become sufficiently available;
- local-bank bonds, divided into interbank negotiable certificates of deposit (IBNCDs), senior bonds, and subordinated bonds.

For bank subordinated bonds, the paper measures spread using yield-to-execution minus the CDB yield at the same maturity-to-execution, because these bonds are callable and are typically redeemed on the call date.

Interpretation. Subordinated bank bonds are the cleanest banking-security outcome because they are junior claims and therefore especially sensitive to changes in bank solvency.

2.8 Local bank sample and outcome variables

For bank real effects, the paper uses two bank samples.

First, it uses a **disclosed-bank sample** of urban and rural commercial banks with quarterly financial statements. This sample is used for outcomes such as:

- net interest margin,
- total assets,
- total loans,
- net profit,
- loans-to-assets,
- loans-to-deposits,
- financial-investments-to-assets.

Second, it uses a broader sample that also includes banks without full disclosure, so it can study whether a bank has been consolidated.

Interpretation. The paper separates intensive-margin outcomes from the exit margin. That makes the banking results much more complete.

2.9 Descriptive background from Figures 2 and 3

The macro background is stark:

- by the end of 2024, total local-government debt rises above 104.6 trillion CNY, or about 77.5% of GDP;
- the interest-coverage ratio implied by land revenue is close to falling below 1;
- local banks become a much larger part of China's banking system, with their asset share rising from 13.4% in 2010 to 29.9% in 2024;
- among listed local banks, LGFV loans account for about 37.86% of corporate loans and 22.42% of total loans by 2024.

Interpretation. These facts explain why LGFV repricing should matter for local banks. The exposure is not marginal; it is central to the business model of many local lenders.

3 Models and empirical specifications (all equations + interpretation)

3.1 Monthly regression for the risky-city spread gap

The paper's first reduced-form object is a monthly rolling panel regression:

$$CreditSpread_{i,c,t} = a + b_t RiskyCityDummy_{c,t} + \sum_k Controls_{k,i,c,t} + \varepsilon_{i,c,t}. \quad (1)$$

Here, i indexes bonds, c cities, and t months. The controls include bond characteristics, LGFV characteristics, and month fixed effects. The paper estimates this in rolling 12-month windows and uses two-way clustered standard errors.

Interpretation. The coefficient b_t is the cleaned-up risky-minus-healthy spread gap at month t . If b_t rises sharply after a policy, risky-city LGFV bonds are repricing upward relative to healthy-city bonds even after controlling for observable bond and issuer differences.

3.2 Expected payoff under default risk

The structural discussion begins with a representative zero-coupon LGFV bond with face value F , maturity T , default intensity $q_{p,t}$, and expected fractional loss given default $l_{p,t}$:

$$E_t^Q[D_{t+T}] = F \left[(1 - q_{p,t}T) + q_{p,t}T(1 - l_{p,t}) \right] = F(1 - q_{p,t}l_{p,t}T). \quad (2)$$

Interpretation. Expected payoff is face value minus expected default loss. In the simplest version, the spread is just compensation for that expected loss.

3.3 From expected loss to the bond credit spread

Let $y_{p,t}$ be the yield to maturity and r the risk-free rate. Then the bond spread is approximately:

$$s_{p,t} \equiv y_{p,t} - r = -\frac{1}{T} \ln \left(\frac{E_t^Q[D_{t+T}]}{F} \right) \approx q_{p,t}l_{p,t}. \quad (3)$$

Interpretation. In the baseline Duffie-Singleton setup, the spread is approximately the annualized expected default loss.

3.4 Adding a liquidity premium

The paper then allows for a liquidity shock that forces investors to sell at a proportional transaction cost $k_{p,t}$, with shock intensity ξ_t :

$$s_{p,t} \approx q_{p,t}l_{p,t} + \xi_t k_{p,t}. \quad (4)$$

Interpretation. Now the spread has two parts: compensation for default loss and compensation for illiquidity.

3.5 Adding central-government bailout probability

Let π_t be the risk-neutral probability that the central government fully bails out the bond in default, and let $L_{p,t}$ be the expected fractional loss *if there is no bailout*. Then:

$$s_{p,t} \approx q_{p,t}L_{p,t}(1 - \pi_t) + \xi_t k_{p,t}. \quad (5)$$

Interpretation. This is the key equation of the paper. Bailout beliefs do not merely shift spreads by an additive constant. They *scale* how much fundamentals matter. When π_t is high, even weak places do not trade at dramatically wider spreads because investors think the center will step in.

3.6 Cross-region spread difference

For risky region r and healthy region h , the cross-region spread gap becomes:

$$s_{r,t} - s_{h,t} \approx (q_{r,t}L_{r,t} - q_{h,t}L_{h,t})(1 - \pi_t) + \xi_t(k_{r,t} - k_{h,t}). \quad (6)$$

Interpretation. This delivers the three mechanisms in one line:

- a **fundamentals mechanism** through qL ,
- a **liquidity mechanism** through $\xi_t(k_r - k_h)$,
- a **bailout mechanism** through the common multiplier $(1 - \pi_t)$.

The paper's main claim is that the third term is the dominant force behind the cycle.

3.7 Regression linking spread sensitivity to fundamentals

To test whether the pricing of fundamentals changes over time, the paper estimates rolling regressions of the form:

$$CreditSpread_{i,p,t} = a + b_t ProvFundamental_{p,t} + Liquidity_{i,p,t} + \sum_k Controls_{k,i,p,t} + \varepsilon_{i,p,t}. \quad (7)$$

The provincial fundamental is alternately debt-to-GDP, log GDP, fiscal self-sufficiency, or lagged municipal-bond spreads.

Interpretation. If b_t rises when the risky-minus-healthy spread gap rises, then the market is pricing the same fundamentals more aggressively. That is exactly what the bailout mechanism predicts.

3.8 Spillover regression to bank subordinated bonds and municipal bonds

After Debt Resolution, the paper studies whether changes in LGFV spreads forecast changes in other bonds:

$$\Delta Spread_{i,c,p,t+\tau} = a + b_\tau \Delta AvgLGFV Spread_{c,p,t} + \sum_k Controls_{k,i,c,p,t} + \alpha_t + \varepsilon_{i,c,p,t+\tau}. \quad (8)$$

Here τ ranges from 0 to 6 months.

Interpretation. If $b_\tau > 0$, then a larger compression in local LGFV spreads today predicts a larger compression in bank subordinated-bond spreads or municipal-bond spreads later. That is direct evidence of pricing spillovers.

3.9 Difference-in-differences for bank outcomes

For bank profitability, size, and portfolio outcomes, the paper estimates:

$$Y_{i,c,t} = a + b(CityRiskyMeasure_c \times Post_t) + \sum_k Controls_{k,i,c,t} + \varepsilon_{i,c,t}. \quad (9)$$

The treatment intensity is continuous: cities with higher pre-policy LGFV spreads are more exposed to the policy shock.

Interpretation. This is a continuous-treatment DID. The coefficient b asks whether banks in riskier cities move more after Debt Resolution than banks in healthier cities.

3.10 Dynamic event study

To test pre-trends and trace the timing, the paper estimates:

$$Y_{i,c,t} = \sum_{k=-4}^{-2} b_k^{lead}(CityRiskyMeasure_c \times D_t^k) + \sum_{k=1}^6 b_k^{lag}(CityRiskyMeasure_c \times D_t^k) + \Gamma' Controls_{i,c,t} + \alpha_t + \alpha_{BankType} + \varepsilon_{i,c,t}, \quad (10)$$

where the quarter just before the policy is the omitted baseline.

Interpretation. Insignificant lead coefficients support the parallel-trends claim. Rising lag coefficients show whether the heterogeneous treatment effect grows over time.

3.11 Within-city heterogeneity by bank LGFV exposure

To show that the mechanism also works *within* a city, the paper estimates a triple-difference specification:

$$Y_{i,c,t} = a + b(CityRiskyMeasure_c \times Post_t \times BankLGFVExposure_{i,c}) + \sum_k Controls_{k,i,c,t} + \varepsilon_{i,c,t}. \quad (11)$$

Interpretation. This is powerful because it compares banks facing the same city-level conditions. If more LGFV-exposed banks inside the same city are hit harder, the story is much more likely to be LGFV-specific rather than just “weak cities do worse.”

3.12 Consolidation probability

Finally, the paper studies bank exit using a probit specification:

$$\Phi^{-1}(BankConsolidatedDummy_{i,c,t}) = a + b(CityRiskyMeasure_c \times Post_t) + \sum_k Controls_{k,i,c,t} + \varepsilon_{i,c,t}. \quad (12)$$

Interpretation. This asks whether Debt Resolution pushes banks in riskier cities closer to the exit margin, measured by subsequent consolidation.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no classical IV here

The paper does *not* rely on a traditional external instrument. Identification comes from policy timing, cross-sectional exposure, dynamic pre-trend tests, and within-city exposure differences.

Interpretation. This is best thought of as a policy-event paper with structural discipline, not an IV paper.

4.2 Why call the cycle government-led?

The cycle is called government-led because the large turning points in spreads line up closely with major policy announcements:

- Directive No.43 only modestly widens the spread gap;
- the 2018 New Regulation creates a much bigger divergence;
- the post-Yongmei strict-regulation period creates the largest divergence;
- the 2023 Debt Resolution package generates an abrupt convergence.

This pattern survives after controlling for bond characteristics, LGFV balance-sheet variables, and month fixed effects, and it is robust to alternative risky-versus-healthy classifications.

Interpretation. The paper’s first identification step is a timing argument: if the cycle were purely macroeconomic, it would be harder to explain why the turning points coincide so neatly with policy news about guarantees and debt resolution.

4.3 Why bailout beliefs instead of only fundamentals?

The key evidence is that the *pricing sensitivity* to fundamentals is itself cyclical. Figure 4 shows that coefficients on debt-to-GDP, GDP, self-sufficiency, and municipal spreads all become much larger in absolute value when the risky-minus-healthy spread gap is widening, and much smaller when the gap compresses after Debt Resolution.

At the same time, the paper shows that direct differences in fundamentals and liquidity between risky and healthy places move in the right direction but are too small to explain most of the spread-gap swing. The residual is attributed to bailout repricing.

Interpretation. If fundamentals alone were doing all the work, then one would mainly expect big changes in fundamentals. Instead, the paper sees big changes in how aggressively the market prices those fundamentals.

4.4 Why Debt Resolution is the cleanest banking shock

Earlier banking events are harder to interpret because they overlap with bank-specific bailout events, especially the Baoshang bank collapse and the write-down of Baoshang subordinated debt. Debt Resolution is cleaner because its transmission operates directly through LGFV repricing.

The chain is:

Debt Resolution → higher perceived LGFV support → lower LGFV spreads → lower expected bank credit losses on LGFV exposure.

Interpretation. This makes the 2023 episode the cleanest setting to ask how LGFV repricing spills into banks.

4.5 Why the bank results are not just weak-city trends

The paper addresses the “poor cities were already deteriorating” alternative in three ways:

- it controls for city and provincial economic indicators such as GDP and leverage;
- event-study graphs show little evidence of pre-trends in the outcome gaps;
- within-city triple-difference results show that more LGFV-exposed banks in the same city are hit harder.

Interpretation. The third point is especially important. City-level weakness cannot easily explain why one bank in a city loses more than another when the main difference is pre-policy LGFV exposure.

4.6 What remains hard to pin down

A few limitations remain:

- bailout probability is not directly observed; it is backed out through the model and the decomposition;
- provincial fundamentals are used because city-level data are weaker, so some local heterogeneity is necessarily missed;
- the paper documents solvency benefits and profitability costs for banks, but it does not perform a full welfare analysis.

Interpretation. The paper makes a strong case that bailout beliefs matter a lot, but the exact numerical bailout probabilities should still be read as model-based calibrations rather than raw observables.

5 Tables and figures

5.1 Figure 1 and Table 3: the government-led LGFV credit cycle

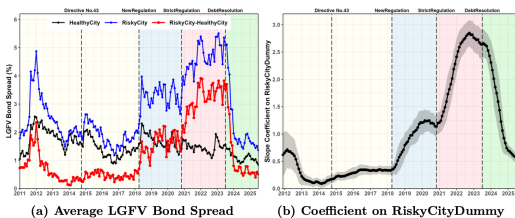


Figure 1. The Government-Led Credit Cycle in LGFV Bonds. Panel (a) plots IssueSize-weighted average credit spreads for LGFV bonds in risky and healthy cities, along with their spread difference. Panel (b) plots monthly slope coefficients from rolling 12-month regressions of LGFV bond credit spreads on *RiskyCityDummy* with control variables. Shaded areas indicate 95% confidence intervals using two-way clustered standard errors. Risky cities are defined as top-tercile cities by city-level IssueSize-weighted average credit spreads of 1-5 year maturity LGFV bonds over the prior 12 months.

Table 3. The Government-Led Credit Cycle in LGFV Bonds. This table reports monthly panel regression (1) for LGFV bonds. The sample period, starting from January 2011 to July 2025, is divided into five phases by four policies: “Directive No.43” in October 2014, “New Regulation” in April 2018, “Strict Regulation” in November 2020, and “Debt Resolution” in July 2023. Credit spreads are in percentages. Reported in parentheses are t-stats using standard errors clustered by bond and month.

Dependent Variable	LGFV Bond Spread					
	Pre Directive No.43 (2011M1-2014M9)	Post Directive No.43 (2014M10-2018M3)	Post NewRegulation (2018M4-2020M10)	Post StrictRegulation (2020M11-2023M6)	Post DebtResolution (2023M7-2025M7)	Recent 1Y (2024M8-2025M7)
	(1)	(2)	(3)	(4)	(5)	(6)
RiskyCityDummy	0.216*** (6.396)	0.321*** (11.734)	1.084*** (15.311)	2.491*** (23.319)	1.064*** (8.754)	0.566*** (12.464)
Rating	0.275*** (9.811)	0.296*** (6.703)	0.487*** (13.284)	0.524*** (12.450)	0.122*** (3.983)	0.060*** (3.907)
Maturity	0.070*** (4.991)	0.027 (1.928)	0.085*** (4.494)	0.046*** (3.053)	0.013 (1.363)	0.010 (1.033)
Age	0.007 (0.591)	-0.057*** (-5.549)	-0.039*** (-2.475)	0.124*** (7.236)	0.058*** (3.758)	0.010 (0.890)
IssueSize	-0.080*** (-2.461)	-0.007 (-0.202)	-0.037 (-0.857)	-0.135*** (-2.969)	-0.097*** (-2.376)	-0.091*** (-2.227)
Exch	-0.218*** (-5.101)	0.261*** (3.898)	0.959*** (3.456)	2.292*** (10.330)	0.739*** (2.647)	-0.029 (-0.183)
Callable	0.361*** (5.288)	2.137*** (11.451)	3.867*** (26.555)	3.222*** (19.890)	2.831*** (25.829)	2.737*** (26.648)
Putable	-0.074 (-1.456)	0.081* (1.820)	-0.124* (-1.846)	-0.205*** (-3.461)	0.368*** (5.968)	0.502*** (10.475)
LnTotalAsset>LastYear	-0.201*** (-4.913)	-0.101*** (-4.300)	-0.228*** (-7.042)	-0.227*** (-6.272)	-0.098*** (-3.468)	-0.148*** (-2.954)
ROA>LastYear	-0.010 (-0.014)	-0.014 (-1.672)	-0.005 (-0.236)	0.018 (0.111)	-0.039* (-1.944)	-0.063*** (-1.366)
Leverage>LastYear	-0.003* (-2.213)	-0.002* (-1.784)	-0.005*** (-2.958)	-0.008*** (-3.537)	-0.006*** (-3.090)	-0.007* (-1.954)
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,644	23,004	16,643	27,691	23,650	10,100
Adjusted R ²	0.534	0.408	0.442	0.522	0.451	0.528

Read: Figure 1 is the central exhibit. Panel (a) plots average LGFV spreads in risky and healthy cities and shows three divergence waves followed by a sharp convergence. Panel (b) shows the same pattern after controls, using the coefficient on *RiskyCityDummy* from rolling regressions. Table 3 then summarizes the phase-by-phase magnitudes: the cleaned risky-minus-healthy spread gap is about 21.6 bps before Directive No.43, about 32.1 bps after Directive No.43, about 108.4 bps after the 2018 New Regulation, about 249.1 bps after Strict Regulation, and then about 106.4 bps on average after Debt Resolution, eventually settling near 56.6 bps in the most recent one-year window.

Economic message. This is the paper’s foundational fact. The spread gap does not move smoothly like an ordinary business-cycle variable; it jumps around policy episodes that change beliefs about central support.

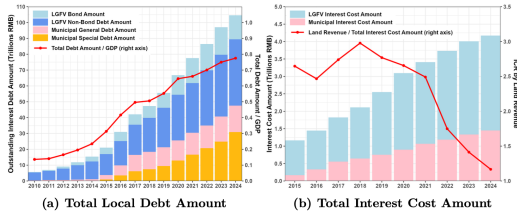


Figure 2. Local Debt Amount and Interest Cost. Panel (a) displays the total outstanding amount of local interest-bearing debt on the left axis, categorized into LGFV bonds, LGFV non-bond debt (primarily loans), municipal general debt, and municipal special debt. The right axis expresses the aggregate local debt as a percentage of GDP. Panel (b) plots total interest costs of LGFV and municipal debts on the left axis, while the right axis shows the interest coverage ratio – calculated as total land revenue divided by total local debt interest costs.

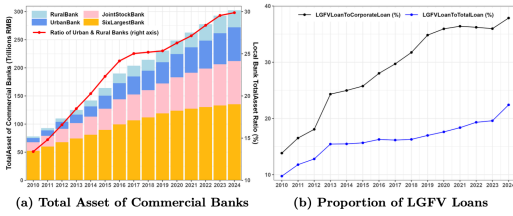


Figure 3. China's Commercial Banks. Panel (a) displays the total asset amount of China's commercial banks on the left axis, categorized into the six largest banks, twelve joint-stock banks, urban commercial banks, and rural commercial banks. The right axis plots the asset ratio of urban and rural commercial banks. Panel (b) presents the ratio of LGFV loan amount to total corporate loan amount (in black line), or to total loan amount (in blue line), among listed urban and rural commercial banks.

5.2 Figures 2 and 3: why the issue matters macroeconomically

Read: Figure 2 shows that local debt explodes through time and that interest coverage based on land revenue falls toward 1. Figure 3 shows that local commercial banks become much larger in the Chinese banking system and that LGFV loans account for a very large share of their loan books.

Economic message. These figures explain why the paper matters. LGFV credit is not a side market. It is large enough that repricing of quasi-sovereign local debt can move the banking system and, through banks, the real economy.

5.3 Figure 4 and Figure 5: the bailout mechanism

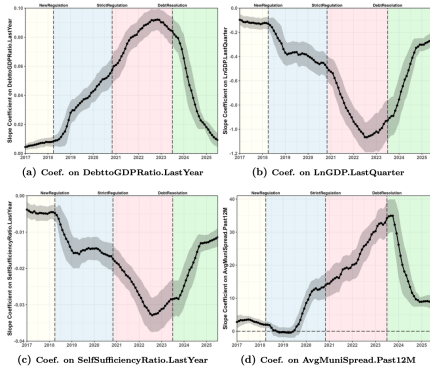


Figure 4. Sensitivity of LGFV Bond Credit Spreads to Fundamental Profiles. Panel (a)-(d) present monthly slope coefficients from rolling 12-month regression (7) of LGFV bond credit spreads on: (a) prior-year provincial debt-to-GDP ratio, (b) prior-quarter provincial log GDP, (c) prior-year provincial fiscal self-sufficiency ratio, and (d) provincial issueSize-weighted average credit spreads of 3-7 year maturity municipal bonds over the prior 12 months. Shaded areas indicate 95% confidence intervals using two-way clustered standard errors. All variables except for $\text{LnGDP}_{\text{LastQuarter}}$ are in percentages.

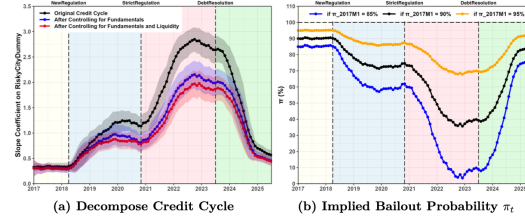


Figure 5. Decompose Credit Cycle. Panel (a) plots monthly slope coefficients from rolling 12-month regressions of LGFV bond credit spreads on RiskyCityDummy_i with different sets of control variables. Shaded areas indicate 95% confidence intervals using two-way clustered standard errors. Panel (b) plots the implied central government bailout probability.

Read: Figure 4 plots the time-varying slope on each fundamentals proxy. The coefficients become much stronger during divergence episodes and then weaken after Debt Resolution. Figure 5 then decomposes the total spread-gap increase from 2017 to the 2022 peak. The paper attributes about 28% of the increase to fundamentals, about 7% to liquidity, and about 65% to the bailout mechanism. Figure 5 also plots an implied bailout-probability path in which support falls sharply through 2022 and then recovers after Debt Resolution.

Economic message. Figure 5 is the intellectual center of the paper. The message is not simply that risky places got worse. It is that the market changed how it mapped those fundamentals into spreads because expected central support changed.

5.4 Figure 6 and Table 4: spillovers to bank subordinated bonds and municipal bonds

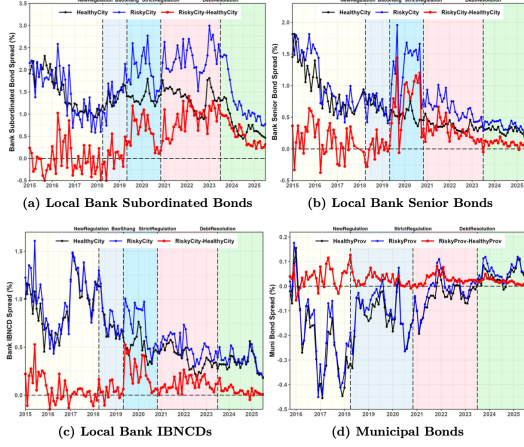


Figure 6. Cross-Regional Spread Difference of Local Bank and Municipal Bonds. These panels plot IssueSize-weighted average credit spreads for (a) local bank subordinated bonds, (b) local bank senior bonds, (c) local bank inter-bank negotiable certificates of deposit (IBNCD), and (d) municipal bonds, separately for risky and healthy regions, along with their spread differences. Risky cities (provinces) are defined uniformly as top-tercile cities (provinces) by city-level (provincial) IssueSize-weighted average credit spreads of 1-5 year maturity LGFV bonds over the prior 12 months.

Table 4. Spillover to Local Bank Subordinated Bonds and Municipal Bonds. This table reports bond-level panel regression (8) which captures the cross-sectional predictability of changes in LGFV bond spreads for subsequent changes in bank subordinated or municipal bond spreads τ months ahead, over the one-year window after “Debt Resolution”. All $\Delta Spread$ are in bps. Reported in parentheses are t-stats using standard errors clustered by bond.

Dependent Variable:	Δ Local Bank Subordinated Bond Spread $_{t,\tau}$						
	$\tau = 0$ (1)	$\tau = 1$ (2)	$\tau = 2$ (3)	$\tau = 3$ (4)	$\tau = 4$ (5)	$\tau = 5$ (6)	$\tau = 6$ (7)
$\Delta CityAvgLGFVSpread_t$	0.068*** (2.935)	0.075*** (2.971)	0.082*** (2.998)	0.084*** (3.083)	0.111*** (4.185)	0.116*** (4.610)	0.126*** (3.720)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,460	2,228	1,981	1,755	1,566	1,373	1,163
Adjusted R ²	0.58	0.56	0.54	0.49	0.43	0.39	0.35

Dependent Variable:	Δ Municipal Bond Spread $_{t,\tau}$						
	$\tau = 0$ (8)	$\tau = 1$ (9)	$\tau = 2$ (10)	$\tau = 3$ (11)	$\tau = 4$ (12)	$\tau = 5$ (13)	$\tau = 6$ (14)
$\Delta ProvAvgLGFVSpread_t$	0.002** (2.258)	0.002** (2.491)	0.003*** (2.989)	0.003*** (3.225)	0.004*** (3.580)	0.004*** (3.499)	0.006*** (4.304)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,546	13,163	11,975	10,704	9,823	8,475	6,957
Adjusted R ²	0.14	0.13	0.14	0.13	0.14	0.13	0.11

Read: Figure 6 shows that spread differences also emerge in local bank subordinated bonds, senior bank bonds, IBNCDs, and municipal bonds. Table 4 then quantifies post-Debt-Resolution spillovers. For local bank subordinated bonds, the coefficient linking current LGFV-spread changes to future bank-spread changes is about 0.068 contemporaneously and rises to about 0.126 at the 6-month horizon. For municipal bonds, the corresponding coefficient rises from about 0.002 to about 0.006.

Economic message. The spillover is real and dynamic. Markets do not absorb the whole implication immediately. The repricing of LGFVs gradually passes through to other local quasi-sovereign and bank liabilities.

5.5 Table 5 and Figure 7(a)–(d): profitability and market-share losses

Read: Table 5 shows that Debt Resolution hurts banks in riskier cities on the real side. A one-standard-deviation increase in pre-policy city risk implies roughly 0.089 percentage points more decline in net interest margin, about 17.02 billion CNY more decline in total assets, about 8.67 billion CNY more decline in total loans, and about 0.135 billion CNY more decline in net profit. Figure 7(a)–(d) shows little evidence of pre-trends and then increasingly negative post-policy effects.

Economic message. This is the paper’s “bad news” result. Lower LGFV spreads make bank balance sheets safer, but they also reduce the profitability of lending to those borrowers and shrink the business volume of banks that relied on them.

5.6 Table 6 and Table 7: within-city exposure and consolidation

Read: Table 6 uses a triple-difference design and shows that even within the same city, banks with higher pre-policy LGFV loan exposure are hit harder after Debt Resolution. Table 7 studies the

Table 5. Real Effects on Local Bank Profitability and Market Shares. This table reports results of DID regression (9) performed on bank-level quarterly panel data. The panel data is constituted of all urban and rural commercial banks with publicly-disclosed financial statements each quarter. The sample period spans from one year prior to the 2023Q3 “Debt Resolution” policy to the most recent available quarter. *NetInterestMargin*, *CityRiskMeasure*, *DebtToGDPRatio.City.LastYear*, *DebtToGDPRatio.Prov.LastYear*, *CoreTier1CapitalAdequacyRatio.LastQuarter* and *ROA.LastQuarter* are in percentages. *TotalAsset*, *TotalLoan*, *NetProfit* and *LnTotalAsset.LastQuarter* are in unit of billion CNY. Reported in parentheses are t-stats using standard errors clustered by bank.

Dependent Variable:	<i>NetInterestMargin</i>		<i>TotalAsset</i>		<i>TotalLoan</i>		<i>NetProfit</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>CityRiskMeasure</i> × Post	-0.050***	-0.041***	-9.918***	-7.809**	-5.196**	-3.977***	-0.087***	-0.062**
	(-4.423)	(-5.930)	(-3.037)	(-2.546)	(-3.167)	(-2.590)	(-3.236)	(-2.464)
<i>CityRiskMeasure</i>	0.131***	0.082**	-15.932**	21.178*	-8.390**	10.782*	0.161***	0.189*
	(9.195)	(4.018)	(-4.624)	(2.841)**	(-4.900)	(2.122)	(5.094)	(2.179)
<i>LnGDP.City.LastYear</i>				23.841**		109.914**		1.559**
				(3.400)		(5.681)		(4.892)
<i>DebtToGDPRatio.City.LastYear</i>				7.037**		3.368***		0.045**
				(2.079)		(3.435)		(2.785)
<i>LnGDP.Prov.LastYear</i>				0.291***		63.370**		-0.786
				(4.394)		(-1.995)		(-1.473)
<i>DebtToGDPRatio.Prov.LastYear</i>				-11.197*		-5.359*		-0.077*
				(2.865)		(-2.402)		(-2.071)
<i>CoreTier1CapitalAdequacyRatio.LastQuarter</i>				-15.748***		-8.784***		-0.137***
				(3.147)		(-4.184)		(-3.878)
<i>ROA.LastQuarter</i>				80.706**		42.806**		1.592***
				(2.246)		(2.303)		(4.680)
<i>LnTotalAsset.LastQuarter</i>				-0.199***				
				(-5.338)				
Quarter & BankType FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,335	4,335	4,734	4,734	4,696	4,696	4,734	4,734
Adjusted R ²	0.21	0.34	0.24	0.40	0.25	0.41	0.15	0.30

Table 6. Real Effects on Local Banks with Heterogeneous LGFV Loan Exposure. This table reports results of DDD regression (11) performed on bank-level quarterly panel data. The panel data is constituted of all urban and rural commercial banks with publicly-disclosed financial statements each quarter. The sample period spans from one year prior to the “Debt Resolution” policy to the most recent available quarter. *NetInterestMargin*, *CityRiskMeasure*, *DebtToGDPRatio.City.LastYear*, *DebtToGDPRatio.Prov.LastYear*, *CoreTier1CapitalAdequacyRatio.LastQuarter* and *ROA.LastQuarter* are in percentages. *TotalAsset*, *TotalLoan*, *NetProfit* and *LnTotalAsset.LastQuarter* are in unit of billion CNY. *BankLGFVExposure* is not in percentages. Reported in parentheses are t-stats using standard errors clustered by bank.

Dependent Variable:	<i>NetInterestMargin</i>		<i>TotalAsset</i>		<i>TotalLoan</i>		<i>NetProfit</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>CityRiskMeasure</i> × Post × <i>BankLGFVExposure</i>	-0.145***	-0.141***	-0.052***	-0.060***	-0.023**	-0.028**	-0.371**	-0.401*
	(-2.921)	(-2.993)	(-2.052)	(-2.501)	(-1.717)	(-1.980)	(-1.742)	(-1.882)
<i>CityRiskMeasure</i> × Post	0.066***	0.060***	0.011	0.014*	0.004	0.005	0.090	0.097
	(3.325)	(3.155)	(1.478)	(1.770)	(0.973)	(1.212)	(1.869)	(1.841)
<i>CityRiskMeasure</i> × <i>BankLGFVExposure</i>	0.296	0.224	-0.027	0.048	-0.013	0.028	0.015	0.376
	(1.547)	(1.284)	(-0.418)	(0.583)	(-0.415)	(0.692)	(0.020)	(0.496)
Post × <i>BankLGFVExposure</i>	0.303**	0.270*	0.305***	0.259**	0.102***	0.127***	1.147*	1.607***
	(2.089)	(1.950)	(2.959)	(3.529)	(2.815)	(3.299)	(1.947)	(2.721)
<i>CityRiskMeasure</i>	-0.134*	-0.087	-0.031*	-0.066	-0.025*	0.003	-0.046*	-0.051
	(-1.721)	(-1.137)	(-1.709)	(-0.181)	(-1.665)	(-0.215)	(-2.398)	(-0.176)
<i>BankLGFVExposure</i>	-0.518	-0.137	0.565***	-0.071	0.257**	-0.070	4.525**	0.283
	(-1.017)	(-0.138)	(2.634)	(-0.310)	(2.307)	(-0.636)	(2.346)	(0.120)
<i>LnGDP.City.LastYear</i>			0.005	0.005		0.004**		0.048*
			(0.081)	(0.394)		(5.299)		(4.811)
<i>DebtToGDPRatio.City.LastYear</i>			0.005	0.008*		0.004**		0.048*
			(1.286)	(2.531)		(2.326)		(1.677)
<i>LnGDP.Prov.LastYear</i>			0.242**	-0.162*		-0.087*		-1.111
			(2.360)	(-1.966)		(-1.960)		(-1.509)
<i>DebtToGDPRatio.Prov.LastYear</i>			0.000	-0.016*		-0.008**		-0.119*
			(0.050)	(-2.216)		(-2.226)		(-1.826)
<i>CoreTier1CapitalAdequacyRatio.LastQuarter</i>			0.010	-0.019*		-0.022**		-0.149*
			(0.453)	(-2.142)		(-2.725)		(-1.821)
<i>ROA.LastQuarter</i>				0.205*		0.103*		3.615***
				(2.409)		(2.405)		(4.472)
<i>LnTotalAsset.LastQuarter</i>				-0.137***				
				(-3.363)				
Quarter & BankType FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,080	2,080	2,090	2,090	2,089	2,089	2,089	2,089
Adjusted R ²	0.11	0.21	0.21	0.41	0.21	0.41	0.14	0.35

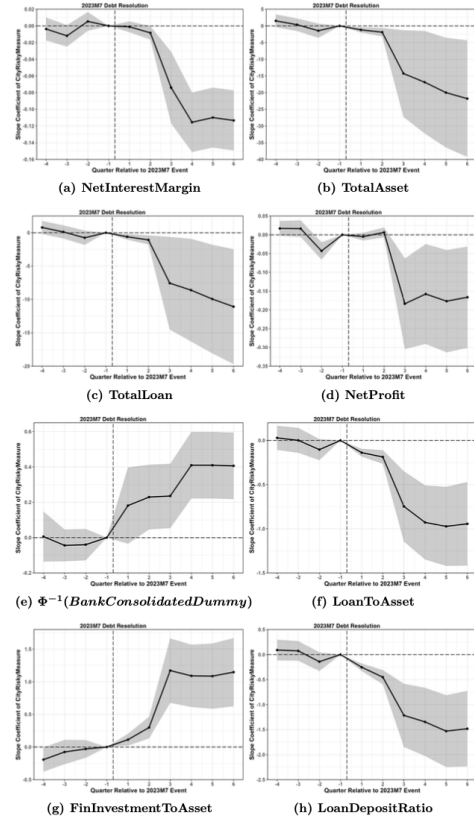


Figure 7. Dynamic Event Study Plots. Each panel plots b_k estimated from panel regression (10) with different dependent variables. *CityRiskMeasure*, *NetInterestMargin*, *LoanToAsset*, *FinInvestmentToAsset*, *FinInvestmentToAsset*, and *LoanDepositRatio* are in percentages. *TotalAsset*, *TotalLoan*, and *NetProfit* are in billion CNY. The shaded area indicates the 95% confidence intervals using standard errors clustered by bank.

Table 7. Real Effects on Local Bank Consolidation Probability. This table reports results of DID Probit regression (12) performed on bank-level quarterly panel data. The panel data is constituted of all urban and rural commercial banks including both institutions with and without publicly disclosed financial statements. The sample period spans from one year prior to the “Debt Resolution” policy to the most recent available quarter. *CityRiskMeasure*, *DebtToGDPRatio.City.LastYear* and *DebtToGDPRatio.Prov.LastYear* are in percentages. Reported in parentheses are z-stats using standard errors clustered by bank. The McFadden’s adjusted pseudo R^2 (typically < 0.4 for discrete outcomes) is computed as $Adj.R^2_{McFadden} = 1 - \frac{\ln L_{empirical} - K}{\ln L_{null}}$, where L denotes the likelihood, and K represents the number of independent variables.

Dependent Variable:	$\Phi^{-1}(BankConsolidatedDummy)$	
	(1)	(2)
<i>CityRiskMeasure</i> × Post	0.270***	0.371***
	(2.983)	(3.400)
<i>CityRiskMeasure</i>	-0.261***	-0.403***
	(-2.786)	(-3.319)
<i>LnGDP.City.LastYear</i>		-0.418***
		(-2.727)
<i>DebtToGDPRatio.City.LastYear</i>		-0.004
		(-0.396)
<i>LnGDP.Prov.LastYear</i>		-0.050
		(-0.236)
<i>DebtToGDPRatio.Prov.LastYear</i>		-0.001
		(-0.059)
Quarter & BankType FE	Yes	Yes
Observations	12,915	12,915
Pseudo R ²	0.08	0.11

exit margin and finds that a one-standard-deviation increase in city risk is associated with about a 0.9% higher increase in the probability of consolidation.

Economic message. These two tables make the mechanism more convincing. The results are not just that “bad cities do badly.” They are also that banks more tied to LGFVs do worse inside the same city, and some are pushed closer to exit.

5.7 Table 8 and Figure 7(e)–(h): operation adjustment

Table 8. Real Effects on Local Bank Operations. This table reports results of DID regression (9) performed on bank-level quarterly panel data. The panel data is constituted of all urban and rural commercial banks with publicly-disclosed financial statements each quarter. The sample period spans from one year prior to the “Debt Resolution” policy to the most recent available quarter. All dependent variables, *CityRiskMeasure*, *ROA.LastQuarter*, *CoreTier1CapitalAdequacyRatio.LastQuarter*, *DebtToGDPRatio.City.LastYear* and *DebtToGDPRatio.Prov.LastYear* are in percentages. Reported in parentheses are t-stats using standard errors clustered by bank.

Dependent Variable:	LoanToAsset (1)	FinInvestmentToAsset (2)	LoanDepositRatio (3)	FinInvestmentToAsset (4)	LoanDepositRatio (5)	FinInvestmentToAsset (6)
CityRiskMeasure × Post	-0.504*** (-5.454)	-0.414*** (-4.922)	0.727*** (5.148)	0.655*** (5.004)	-0.901*** (-5.838)	-0.749*** (-5.205)
CityRiskMeasure	0.493*** (4.001)	0.284 (1.326)	-0.880*** (-4.732)	-0.354 (-1.317)	0.614*** (3.120)	0.667** (2.006)
LnTotalAsset.LastQuarter		-2.229*** (-3.350)		2.550*** (5.971)		-0.143 (-0.252)
CoreTier1CapitalAdequacyRatio.LastQuarter		-0.614*** (-4.144)		0.943*** (2.589)		-0.315 (-1.141)
NetProfitToTotalAsset.LastQuarter		4.411*** (4.709)		-2.314** (-2.510)		7.237*** (4.832)
LnGDP.City.LastYear		1.392*** (2.631)		-1.986*** (-3.575)		2.902*** (3.728)
DebtToGDPRatio.City.LastYear		-0.005 (-0.148)		-0.045 (-1.290)		0.005 (0.093)
LnGDP.Prov.LastYear		2.851*** (3.617)		-1.170 (-1.138)		2.459** (2.058)
DebtToGDPRatio.Prov.LastYear		0.159*** (2.799)		-0.129* (-1.847)		0.236*** (2.846)
Quarter & BankType FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,715	4,715	4,424	4,424	4,663	4,663
Adjusted R ²	0.07	0.18	0.08	0.15	0.04	0.15

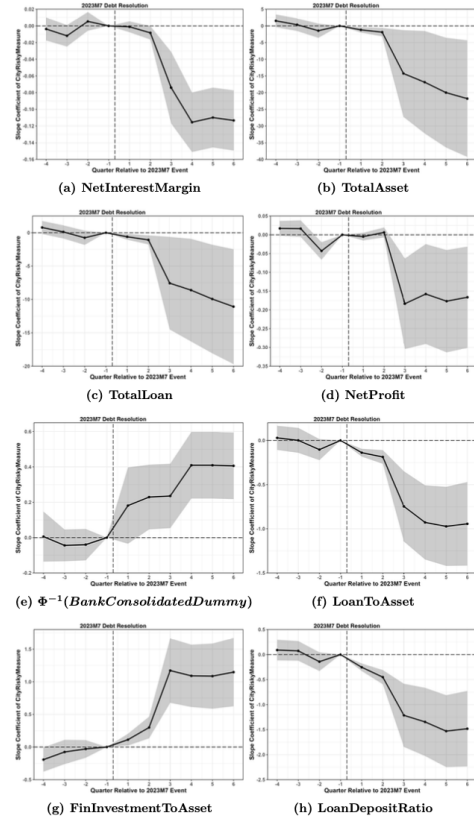


Figure 7. Dynamic Event Study Plots. Each panel plots b_t estimated from panel regression (10) with different dependent variables. *CityRiskMeasure*, *NetInterestMargin*, *LoanToAsset*, *FinInvestmentToAsset*, *FinInvestmentToAsset*, and *LoanDepositRatio* are in percentages. *TotalAsset*, *TotalLoan*, and *NetProfit* are in billion CNY. The shaded area indicates the 95% confidence intervals using standard errors clustered by bank.

Read: Table 8 shows that banks in riskier cities respond by changing their business mix. A one-standard-deviation increase in pre-policy city risk is associated with about 0.9 percentage points more decline in loans-to-assets, about 1.4 percentage points more decline in loans-to-deposits, and about 1.6 percentage points more increase in financial-investments-to-assets. Figure 7(e)–(h) again shows weak pre-trends and growing post-policy effects.

Economic message. The banks do not just passively absorb the shock. They rebalance away from traditional lending and toward financial assets. That is a structural change in local intermediation.

6 Short summary

- The paper studies how China's changing stance toward local-government hidden debt is priced in LGFV bonds.
- LGFV bonds are the ideal asset for this question because they are quasi-fiscal liabilities with strong links to local governments and ambiguous bailout status.
- The main fact is a government-led credit cycle: risky-city LGFV spreads diverge from healthy-city spreads after the 2018 and 2020 policy turns, then compress sharply after the 2023 Debt Resolution package.
- A simple credit-risk model says spreads depend on fundamentals, liquidity, and bailout probability.
- Empirically, the paper argues that the biggest driver of the cycle is changing bailout beliefs, not just changing fundamentals or liquidity.
- The paper estimates that from 2017 to the 2022 peak, about 65% of the spread-gap increase comes from bailout repricing.
- Debt Resolution narrows LGFV spreads and this spills over into lower subordinated-bond spreads for local banks and lower municipal-bond spreads.
- But the policy also depresses profitability and market share for banks in riskier cities, raises their consolidation risk, and pushes them toward financial investments.
- The overall message is that government support beliefs can create a large credit cycle in quasi-sovereign debt and reshape the structure of local banking.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that China's LGFV market experienced a large, policy-driven credit cycle over the last decade, and that this cycle is best understood as a cycle in perceived central-government support rather than simply a cycle in regional fundamentals.

7.2 Economic intuition

When investors think the center will rescue distressed local debts, risky and healthy cities do not look all that different in bond spreads because the downside is partially socialized. When the market starts to believe that local governments may be forced to bear their own hidden debts, the same underlying fundamentals suddenly matter much more. That is why risky-city spreads explode relative to healthy-city spreads. Debt Resolution partially reverses this logic by restoring faith in support.

7.3 What the paper adds

The paper contributes two linked insights. First, it shows how a quasi-sovereign credit cycle can be *government-led*. Second, it shows that resolving local debt stress is not a free lunch for banks: the same policy that lowers expected credit loss can also compress lending margins and accelerate concentration in local banking.

7.4 Final takeaway

The deepest message is that credit markets price not only cash flows and collateral, but also political willingness to stand behind debt. In China's LGFV market, that political backstop moved enough to generate a full credit cycle, and the resulting repricing spilled directly into the health, size, and business models of local commercial banks.